

Calendars

2024-10-07

```
library(rjd3toolkit)

##
## Attaching package: 'rjd3toolkit'

## The following objects are masked from 'package:stats':
##
##      aggregate, mad

library(rjd3tramoseats)
library(rjd3providers)
data<-file.path(rprojroot::find_root(rprojroot::is_rstudio_project ), 'Data')
set_spreadsheet_paths(data)
xs<-spreadsheet_data("Belgium_XM.xlsx", 1)$series
ms<-spreadsheet_data("Belgium_XM.xlsx", 2)$series
s<-ABS$X0.2.20.10.M
```

Calendars

Trading days

```
td_raw<-td(s=s, contrasts=FALSE)
colnames(td_raw)<-c("Sundays", "Mondays", "Tuesdays", "Wednesdays", "Thursdays", "Fridays", "Saturdays")
print(window(td_raw, start=c(2000,1), end=c(2001,12)))
```

	Sundays	Mondays	Tuesdays	Wednesdays	Thursdays	Fridays	Saturdays
## Jan 2000	5	5	4	4	4	4	5
## Feb 2000	4	4	5	4	4	4	4
## Mar 2000	4	4	4	5	5	5	4
## Apr 2000	5	4	4	4	4	4	5
## May 2000	4	5	5	5	4	4	4
## Jun 2000	4	4	4	4	5	5	4
## Jul 2000	5	5	4	4	4	4	5
## Aug 2000	4	4	5	5	5	4	4
## Sep 2000	4	4	4	4	4	5	5
## Oct 2000	5	5	5	4	4	4	4
## Nov 2000	4	4	4	5	5	4	4
## Dec 2000	5	4	4	4	4	5	5
## Jan 2001	4	5	5	5	4	4	4
## Feb 2001	4	4	4	4	4	4	4
## Mar 2001	4	4	4	4	5	5	5
## Apr 2001	5	5	4	4	4	4	4
## May 2001	4	4	5	5	5	4	4
## Jun 2001	4	4	4	4	4	5	5
## Jul 2001	5	5	5	4	4	4	4
## Aug 2001	4	4	4	5	5	5	4

```
## Sep 2001      5      4      4      4      4      4      5
## Oct 2001      4      5      5      5      4      4      4
## Nov 2001      4      4      4      4      5      5      4
## Dec 2001      5      5      4      4      4      4      5
```

```
model1<-sarima_estimate(log(s), order=c(0,1,1), seasonal=list(order=c(0,1,1), period=12), xreg = td_raw)
print(model1)
```

```
## SARIMA model: (0,1,1) (0,1,1) [12]
```

```
##
```

```
## Coefficients
```

```
## theta(1) btheta(1)
```

```
## -0.6351 -0.6487
```

```
##
```

```
## Regression model:
```

```
##      Sundays      Mondays      Tuesdays      Wednesdays      Thursdays      Fridays      Saturdays
```

```
##      0.01990      0.02856      0.03207      0.03278      0.03914      0.03843      0.03365
```

```
##
```

```
## For a more detailed output, use the 'summary()' function.
```

```
cat("\nAverage effect:", mean(model1$b), "\n\n")
```

```
##
```

```
## Average effect: 0.03207658
```

```
lp<-lp_variable(s=s, type = "LengthOfPeriod")
```

```
td2<-cbind(td_raw[,-1]-td_raw[,1], lp)
```

```
colnames(td2)<-c("Mondays", "Tuesdays", "Wednesdays", "Thursdays", "Fridays", "Saturdays", "lp")
```

```
print(window(td2, start=c(2000,1), end=c(2001,12)))
```

```
##      Mondays Tuesdays Wednesdays Thursdays Fridays Saturdays      lp
## Jan 2000      0      -1      -1      -1      -1      0 0.5625
## Feb 2000      0       1       0       0       0      0 -1.4375
## Mar 2000      0       0       1       1       1      0 0.5625
## Apr 2000     -1     -1     -1     -1     -1      0 -0.4375
## May 2000      1       1       1       0       0      0 0.5625
## Jun 2000      0       0       0       1       1      0 -0.4375
## Jul 2000      0     -1     -1     -1     -1      0 0.5625
## Aug 2000      0       1       1       1       0      0 0.5625
## Sep 2000      0       0       0       0       1      1 -0.4375
## Oct 2000      0       0     -1     -1     -1     -1 0.5625
## Nov 2000      0       0       1       1       0      0 -0.4375
## Dec 2000     -1     -1     -1     -1       0      0 0.5625
## Jan 2001      1       1       1       0       0      0 0.5625
## Feb 2001      0       0       0       0       0      0 -2.4375
## Mar 2001      0       0       0       1       1      1 0.5625
## Apr 2001      0     -1     -1     -1     -1     -1 -0.4375
## May 2001      0       1       1       1       0      0 0.5625
## Jun 2001      0       0       0       0       1      1 -0.4375
## Jul 2001      0       0     -1     -1     -1     -1 0.5625
## Aug 2001      0       0       1       1       1      0 0.5625
## Sep 2001     -1     -1     -1     -1     -1      0 -0.4375
## Oct 2001      1       1       1       0       0      0 0.5625
## Nov 2001      0       0       0       1       1      0 -0.4375
## Dec 2001      0     -1     -1     -1     -1      0 0.5625
```

```
model2<-sarima_estimate(log(s), order=c(0,1,1), seasonal=list(order=c(0,1,1), period=12), xreg = td2)
print(model2)
```

```
## SARIMA model: (0,1,1) (0,1,1) [12]
##
## Coefficients
## theta(1) btheta(1)
## -0.6351 -0.6487
##
## Regression model:
## Mondays Tuesdays Wednesdays Thursdays Fridays Saturdays lp
## -3.515e-03 -6.452e-06 7.017e-04 7.066e-03 6.358e-03 1.569e-03 3.208e-02
##
## For a more detailed output, use the 'summary()' function.
```

```
print(model1$b[-1]-model2$b[-7])
```

```
## Mondays Tuesdays Wednesdays Thursdays Fridays Saturdays
## 0.03207658 0.03207658 0.03207658 0.03207658 0.03207658 0.03207658
```

```
print(-sum(model2$b[1:6]))
```

```
## [1] -0.01217378
```

```
lp<-lp_variable(s=s, type = "LeapYear")
td3<-cbind(td_raw[,-5]-td_raw[,5], lp)
colnames(td3)<-c("Sundays", "Mondays", "Tuesdays", "Wednesdays", "Fridays", "Saturdays", "lp")
print(window(td3, start=c(2000,1), end=c(2001,12)))
```

```
##      Sundays Mondays Tuesdays Wednesdays Fridays Saturdays    lp
## Jan 2000      1      1      0      0      0      1 0.00
## Feb 2000      0      0      1      0      0      0 0.75
## Mar 2000     -1     -1     -1      0      0     -1 0.00
## Apr 2000      1      0      0      0      0      1 0.00
## May 2000      0      1      1      1      0      0 0.00
## Jun 2000     -1     -1     -1     -1      0     -1 0.00
## Jul 2000      1      1      0      0      0      1 0.00
## Aug 2000     -1     -1      0      0     -1     -1 0.00
## Sep 2000      0      0      0      0      1      1 0.00
## Oct 2000      1      1      1      0      0      0 0.00
## Nov 2000     -1     -1     -1      0     -1     -1 0.00
## Dec 2000      1      0      0      0      1      1 0.00
## Jan 2001      0      1      1      1      0      0 0.00
## Feb 2001      0      0      0      0      0      0 -0.25
## Mar 2001     -1     -1     -1     -1      0      0 0.00
## Apr 2001      1      1      0      0      0      0 0.00
## May 2001     -1     -1      0      0     -1     -1 0.00
## Jun 2001      0      0      0      0      1      1 0.00
## Jul 2001      1      1      1      0      0      0 0.00
## Aug 2001     -1     -1     -1      0      0     -1 0.00
## Sep 2001      1      0      0      0      0      1 0.00
## Oct 2001      0      1      1      1      0      0 0.00
## Nov 2001     -1     -1     -1     -1      0     -1 0.00
## Dec 2001      1      1      0      0      0      1 0.00
```

```
model3<-sarima_estimate(log(s), order=c(0,1,1), seasonal=list(order=c(0,1,1), period=12), xreg = td3)
print(model3)
```

```
## SARIMA model: (0,1,1) (0,1,1) [12]
##
## Coefficients
## theta(1) btheta(1)
## -0.6351 -0.6487
##
## Regression model:
## Sundays Mondays Tuesdays Wednesdays Fridays Saturdays lp
## -1.217e-02 -3.515e-03 -6.452e-06 7.017e-04 6.358e-03 1.569e-03 3.208e-02
##
## For a more detailed output, use the 'summary()' function.
```

```
print(model1$b-mean(model1$b))
```

```
## Sundays Mondays Tuesdays Wednesdays Thursdays
## -1.217378e-02 -3.514939e-03 -6.451643e-06 7.016572e-04 7.065837e-03
## Fridays Saturdays
## 6.358293e-03 1.569382e-03
```

```
print(model2$b)
```

```
## Mondays Tuesdays Wednesdays Thursdays Fridays
## -3.514939e-03 -6.451643e-06 7.016572e-04 7.065837e-03 6.358293e-03
## Saturdays lp
## 1.569382e-03 3.207658e-02
```

```
print(model3$b)
```

```
## Sundays Mondays Tuesdays Wednesdays Fridays
## -1.217378e-02 -3.514939e-03 -6.451639e-06 7.016572e-04 6.358293e-03
## Saturdays lp
## 1.569382e-03 3.207658e-02
```

```
print(-sum(model3$b[1:6]))
```

```
## [1] 0.007065837
```